

Racial/Ethnic Patterns in Lifetime and Age-Conditional Risk Estimates for Selected Cancers

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BACKGROUND. Estimates of the probability of developing or dying from cancer, either over a lifetime or over a specified number of years, are useful summary measures of the burden of cancer in a population.

METHODS. The authors used publicly available DevCan software and new, detailed, racial/ethnic data bases that were developed in the Surveillance Research Program of the National Cancer Institute to produce risk estimates for selected major cancers among American Indian/Aleut/Eskimo, black, Chinese, Filipino, native Hawaiian, Japanese, white (total, non-Hispanic), and Hispanic populations.

RESULTS. Japanese and non-Hispanic white men had the highest lifetime risk for developing cancer (47.94% and 47.41%, respectively), and the American Indian/Eskimo/Aleut population (excluding Alaska) had the lowest lifetime risk among men (24.30%). Among women, white and American Indian/Eskimo/Aleut (in Alaska) populations had higher lifetime risks than Japanese women, whereas American Indian/Eskimo/Aleut (excluding Alaska) women had the lowest risk. The age-conditional probabilities of developing cancer within the next 10 years among men and women age 60 years and the lifetime probabilities of dying from cancer also were reported by racial/ethnic group.

CONCLUSIONS. Racial/ethnic disparities in the lifetime risk of cancer may be because of differences in cancer incidence rates, but they also may reflect differential mortality rates from causes other than the cancer of interest. Furthermore, because cross-sectional incidence and mortality rates are used in calculating the DevCan lifetime risk estimates, results must be interpreted with caution when events, such as the widespread and rapid implementation of a new screening test, are known to have influenced disease rates. *Cancer* 2006;106:670–82.

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KEYWORDS: cancer, health disparities, risk, incidence, mortality, race, ethnic groups, Surveillance, Epidemiology, and End Results Program.

Lifetime and age-conditional estimates of cancer risk are used widely to summarize the cancer experience of the general population and to compare cancer risks among population subgroups.^{1–8} The DevCan software,⁹ which was developed within the Surveillance Program of the National Cancer Institute, calculates the probability of developing or dying from cancer by using incidence data from registries that participate in the Surveillance, Epidemiology, and End Results (SEER) Program; mortality data compiled by the National Center for Health Statistics; and population estimates from the United States Bureau of the Census. Risk estimates of this type represent a measure of the population burden of cancer, because they integrate the chance of developing cancer and the chance of dying of other causes while cancer-free.

In this article, we present lifetime probabilities of developing or

dying from selected major cancers and age-conditional probabilities of developing cancer for several racial/ethnic groups. Previous versions of the DevCan software produced risk estimates only for whites, blacks, and all races combined. Now, the racial/ethnic data have been expanded to enable cancer risk comparisons among American Indian/Aleut/Eskimo, black, Chinese, Filipino, native Hawaiian, Japanese, all Asian and Pacific Islander groups combined, white (total, non-Hispanic), and Hispanic populations. Another recent addition to the DevCan software is the computation of confidence intervals associated with the risk estimates.¹⁰ The current report demonstrates these new features of DevCan, interprets major findings, and hopefully will prompt cancer researchers and health planners to access this publicly available resource.

MATERIALS AND METHODS

The results presented in this article were generated by using version 5.2 of the DevCan software and special SEER data files,^{11–14} which were based on the 1990 Census, because the population estimates needed to calculate disease rates for several detailed racial/ethnic groups are available only from decennial Census data. The DevCan risk measures provide a new assessment of the cancer burden experienced by these racial/ethnic groups and are a useful supplement to the incidence and mortality rate comparisons published in an earlier report.¹⁵

To derive the probabilities of developing cancer for the detailed racial/ethnic groups, incidence rates were based on cancer diagnoses during 1988–1992 from the following SEER reporting areas: metropolitan Atlanta and 10 nearby rural Georgia counties; Detroit, Los Angeles, San Francisco/Oakland, San Jose/Monterey, and Seattle/Puget Sound; and the states of Alaska (American Indian/Aleut/Eskimo populations only), Connecticut, Hawaii, Iowa, New Mexico, and Utah. The diagnosis years were centered on 1990, so that they would be most compatible with the detailed racial/ethnic population data from the decennial United States Census. The American Indian/Aleut/Eskimo population in the United States consists of over 500 tribal groups with diverse cultures and health behaviors,¹⁶ and regional variations in their cancer patterns have been noted.^{15,17} Therefore, risks for developing cancer in the American Indian/Aleut/Eskimo population are presented separately for the state of Alaska (referred to hereinafter as “Alaska Natives”) and for the remaining SEER areas (referred to hereinafter as “American Indians,” because the Alaska Native component in other geographic areas is small). Incidence data from the state of Connecticut are not in-

cluded in the Hispanic and non-Hispanic white calculations because of the large proportion of patients in that registry for whom Hispanic ethnicity is unknown. When calculating the probabilities of developing cancer, mortality data are based on the same SEER regions that were included in the incidence rate calculations. When calculating the probabilities of dying from cancer, the mortality data are based on all United States deaths for 1988–1992, with the following exceptions. Hispanic and non-Hispanic white mortality data are limited to deaths for 1989–1991, because many states had incomplete Hispanic data before 1989.¹⁸ For the same reason, this calendar year restriction was also applied to the Hispanic and non-Hispanic white mortality data associated with the probability of developing cancer calculations. Furthermore, the states of Connecticut, Louisiana, Maine, Maryland, Mississippi, New Hampshire, New York, North Dakota, Oklahoma, Vermont, and Virginia and the District of Columbia are excluded from the Hispanic mortality calculations because of high proportions of patients with unknown Hispanic ethnicity.¹⁹

The DevCan software program uses life-table methods based on cross-sectional incidence, mortality, and population data for 19 age categories (ages 0 yrs, 1–4 yrs, 5–9 yrs, 10–14 yrs,... 80–84 yrs, and 85 yrs or older) to compute the lifetime and age-conditional probability of developing or dying from cancer along with the associated 95% confidence interval (95% CI).^{3,10,20} On the basis of user selections, the program uses the incidence of first cancers of a specified type and mortality from all causes other than the specified cancer as inputs. A life table is constructed for a hypothetical cancer-free cohort of individuals who are exposed to the appropriate gender-specific and race/ethnic-specific, cross-sectional rates as they age. The number of cancer-free individuals at the start of a subsequent age interval is computed by subtracting the number who develop cancer and the number who die of other causes from the number of cancer-free individuals at the start of the interval. For these computations, it is assumed that mortality from other causes among the cancer-free group can be approximated by mortality from other causes in the total population. Incidence rates calculated for the total population are adjusted upward to reflect the risk for a cancer-free population (i.e., individuals living with cancer are removed from the denominator).²¹

Because the lifetime risk measure is a function of both age-specific incidence rates for a given cancer and mortality rates for all causes other than the cancer, two racial/ethnic groups may have identical incidence rates but very different lifetime risk estimates. Thus, in the tables, we compare lifetime risks, which

are measures of the population burden of cancer, with age-adjusted rates, which standardize the number alive and at risk at each age across racial/ethnic groups. The age-adjusted incidence rates are calculated in the same manner as those published annually by the National Cancer Institute and other organizations by including all primary invasive cancers that occur in the covered populations.²² It should be noted that lifetime risk estimates do not assume any particular lifespan. Rather, all possible life spans are included in the calculation. The lifetime risk is computed as the probability of living to the beginning of an age interval multiplied by the probability of developing cancer during that age interval, summed over all age intervals. The DevCan risk estimate that approximately 14% (or 1 in 7) white women will develop breast cancer over their lifetime⁷ implies that 6 of 7 women will die of other causes without ever having been diagnosed with breast cancer. For example, some of those 6 women may die at age 100 years without ever developing breast cancer, whereas others may not develop breast cancer, because they will die at a younger age from some other cause. Lifetime risk estimates apply only to a cohort of newborns and, as calculated in DevCan, are based on an assumption that the cross-sectional incidence and mortality rates will remain constant over a lifetime. It should be noted, however, that large changes in the incidence rates during a specified period, such as an increase associated with the widespread and rapid implementation of a new screening test, will have a direct impact on the derived lifetime risk estimates. For this reason, age-conditional risk estimates over a shorter interval also are produced by DevCan and may be of greater utility when communicating risk to the public. We present probabilities of developing selected cancers within the next 10 years for men and women age 60 years by racial/ethnic group to demonstrate this feature.

RESULTS

Probability of Developing Cancer

Lifetime probabilities of developing cancer are shown in Tables 1 and 2 for women and men, respectively, by racial/ethnic group for all cancers combined and for selected cancers. The average annual, age-adjusted cancer incidence rates for each gender and racial/ethnic group also are included in the tables for reference. The lifetime risk of developing cancer, for all cancers combined, generally is lower among women compared with men. Alaska Native women are an exception: Their risks for all cancers combined and for colorectal cancer are higher compared with their male counterparts.

Japanese men experience the highest lifetime risk for developing cancer (47.94%; 95% CI, 46.24–49.70%) among all racial/ethnic groups, but their risk is only

slightly higher than that for non-Hispanic white men, and the 95% CIs for these 2 groups overlap. Among the specific cancers shown, Japanese men have the highest lifetime risk for colorectal cancer (9.71%; 95% CI, 9.03–10.44%) and are among the groups with higher risks for cancers of the prostate and urinary bladder, although they have a relatively low lifetime risk for lung cancer. Non-Hispanic white men have the second highest lifetime probability of developing cancer, with high risks for cancers of the prostate, urinary bladder, and lung and bronchus. Although native Hawaiian and black men have the highest lifetime risks for lung cancer, they experience relatively low risks for other selected cancers and appear relatively low in the ranking for all cancers combined. Other male Asian populations show mid-level risks for developing cancer. American Indian and Alaska Native men generally have among the lowest lifetime risks for developing cancer overall and for most of the specific cancers shown. An exception is the relatively high risk for colorectal cancer among Alaska Native men. The age-adjusted incidence rate for colorectal cancer in Alaska Native men is also the highest among all of the racial/ethnic groups.

The racial/ethnic pattern in the lifetime risks of developing cancer among women differs somewhat from that for men. For all cancers combined, the lifetime risks among white women and Alaska Native women exceed the risk in Japanese women. Non-Hispanic white women have the highest lifetime risks for cancers of the breast and the corpus and the uterus, whereas the risks for Alaska Native women are highest for cancers of the lung and the colon and rectum. American Indian women have the lowest lifetime risks for all cancers combined and for each of the specific cancers shown, with the exception of cancer of the corpus and uterus, for which only Alaska Native women have a lower estimated risk (based on few patients, however, and broad, overlapping 95% CIs). The incidence of cancer of the gallbladder, however, is known to be high among American Indian women and, although it is not shown in the tables, their lifetime risk for developing this cancer (1.01%; 95% CI, 0.60–1.70%) is more than twice as high as the risk for the next highest racial/ethnic group.

We ranked the lifetime risks and age-adjusted rates across the racial/ethnic groups to facilitate comparisons of the impact that mortality from other causes has on the population burden of a particular cancer. The racial/ethnic rankings of lifetime risks of developing cancer tend to follow the rankings of age-adjusted incidence rates; however, there are some notable exceptions. Black men have the highest age-

TABLE 1
Lifetime Risk (Percentage) among Women of Developing Invasive Cancer with Age-Adjusted Incidence Rates by Disease Type and Racial/Ethnic Group: Surveillance, Epidemiology, and End Results Areas, 1988–1992

Disease type/group	Lifetime risk of being diagnosed				Age-adjusted incidence			
	Percent	LL (%)	UL (%)	Rank	Rate	LL	UL	Rank
All invasive cancers								
White (total)	39.48	39.32	39.65	1	423.82	422.28	425.36	3
Non-Hispanic white ^a	39.42	39.23	39.61	2	432.09	430.32	433.87	2
Alaska Native ^b	37.32	33.15	42.05	3	449.19	408.42	493.97	1
Japanese	35.79	34.50	37.13	4	301.86	293.21	310.75	7
Hawaiian	34.98	32.52	37.67	5	408.61	386.52	431.86	4
Chinese	34.98	33.43	36.60	6	275.75	267.24	284.50	10
Asian/Pacific Islander	34.94	34.17	35.74	7	295.78	291.32	300.29	8
Hispanic ^a	34.17	33.46	34.90	8	302.10	297.86	306.40	6
Filipina	34.12	31.86	36.58	9	279.96	270.63	289.68	9
Black	31.70	31.26	32.15	10	398.77	394.11	403.48	4
Amer Indian (excl AK) ^c	20.31	18.36	22.50	11	189.31	176.87	202.54	11
All races	38.34	38.19	38.48		413.16	411.78	414.55	
Colon and rectum								
Alaska Native ^b	9.24	7.13	11.98	1	93.62	74.20	117.54	1
Japanese	7.30	6.71	7.93	2	49.30	45.90	52.95	5
Chinese	7.00	6.30	7.78	3	44.75	41.26	48.48	6
Asian/Pacific Islander	6.05	5.72	6.41	4	40.33	38.63	42.10	7
White (total)	6.03	5.96	6.09	5	49.77	49.26	50.29	4
Non-Hispanic white ^a	6.00	5.93	6.07	6	50.31	49.73	50.90	3
Black	5.38	5.20	5.57	7	57.64	55.84	59.49	2
Hispanic ^a	4.74	4.46	5.04	8	31.84	30.39	33.34	9
Hawaiian	4.34	3.44	5.51	9	39.95	32.98	48.20	8
Filipina	4.25	3.45	5.28	10	26.76	23.71	30.26	10
Amer Indian (excl AK) ^c	2.17	1.64	2.96	11	19.54	15.57	24.36	11
All races	5.97	5.91	6.03		50.04	49.56	50.52	
Lung and bronchus								
Alaska Native ^b	6.45	4.83	8.66	1	66.57	51.04	86.51	1
Non-Hispanic white ^a	5.69	5.62	5.76	2	51.46	50.86	52.07	3
White (total)	5.51	5.45	5.57	3	49.23	48.71	49.75	5
Hawaiian	5.50	4.58	6.65	4	54.24	46.25	63.48	2
Chinese	5.40	4.81	6.06	5	34.04	31.03	37.30	6
Black	4.65	4.49	4.81	6	51.23	49.57	52.94	4
Asian/Pacific Islander	3.99	3.73	4.27	7	27.25	25.87	28.69	7
Filipina	3.46	2.76	4.39	8	22.81	20.11	25.94	9
Hispanic ^a	3.45	3.24	3.68	9	24.71	23.46	26.02	8
Japanese	2.76	2.42	3.16	10	18.88	16.84	21.18	10
Amer Indian (excl AK) ^c	2.16	1.63	2.94	11	18.39	14.63	22.99	11
All races	5.27	5.22	5.33		47.66	47.19	48.13	
Breast								
Non-Hispanic white ^a	13.80	13.69	13.90	1	138.98	137.96	140.01	1
White (total)	13.53	13.44	13.63	2	134.91	134.03	135.80	2
Hawaiian	11.97	10.65	13.52	3	131.47	119.23	144.88	3
Japanese	10.70	10.11	11.32	4	99.28	94.33	104.47	6
Black	9.55	9.33	9.78	5	115.42	112.94	117.95	4
Filipina	9.35	8.54	10.33	6	90.46	85.55	95.77	7
Hispanic ^a	9.27	8.97	9.60	7	84.76	82.58	87.01	8
Asian/Pacific Islander	8.89	8.59	9.22	8	83.69	81.42	86.03	9
Alaska Native ^b	8.34	6.73	10.48	9	101.75	84.05	123.44	5
Chinese	7.90	7.28	8.58	10	69.18	65.06	73.54	10
Amer Indian (excl AK) ^c	4.35	3.68	5.23	11	44.70	39.07	51.09	11
All races	12.85	12.76	12.93		129.65	128.87	130.44	
Corpus and uterus								
Non-Hispanic white ^a	2.87	2.82	2.92	1	26.62	26.19	27.07	2
White (total)	2.80	2.76	2.84	2	25.86	25.48	26.25	3
Hawaiian	2.60	2.05	3.40	3	28.83	23.31	35.57	1
Japanese	1.92	1.68	2.20	4	17.60	15.56	19.90	4
Hispanic ^a	1.91	1.77	2.06	5	16.17	15.22	17.19	6
Asian/Pacific Islander	1.64	1.52	1.79	6	15.17	14.21	16.18	7
Black	1.59	1.50	1.69	7	17.08	16.13	18.08	5
Filipina	1.50	1.22	2.01	8	14.35	12.47	16.65	8
Chinese	1.47	1.24	1.77	9	13.14	11.41	15.11	9
Amer Indian (excl AK) ^c	0.94	0.62	1.53	10	8.79	6.40	11.99	11
Alaska Native ^b	0.77	0.37	1.86	11	9.39	4.70	18.94	10
All races	2.61	2.57	2.64		24.33	23.99	24.67	

LL: lower limit of 95% confidence interval; UL: upper limit of 95% confidence interval.

^a Based on data for all Surveillance, Epidemiology, and End Results (SEER) areas except the state of Connecticut, which has a high proportion of patients for which Hispanic ethnicity is unknown.

^b Alaska Native: American Indian/Aleut/Eskimo residents of Alaska at the time of diagnosis.

^c Amer Indian (excl AK): American Indian/Aleut/Eskimo residents of SEER areas, excluding the state of Alaska.

TABLE 2
Lifetime Risk (Percentage) among Men of Developing Invasive Cancer with Age-Adjusted Incidence Rates by Disease Type and Racial/Ethnic Group: Surveillance, Epidemiology, and End Results Areas, 1988–1992

Disease type/group	Lifetime risk of being diagnosed				Age-adjusted incidence			
	Percent	LL (%)	UL (%)	Rank	Rate	LL	UL	Rank
All invasive cancers								
Japanese	47.94	46.24	49.70	1	427.06	415.28	439.15	6
Non-Hispanic white ^a	47.41	47.17	47.64	2	619.94	617.48	622.40	2
White (total)	46.48	46.29	46.68	3	601.95	599.79	604.11	3
Asian/Pacific Islander	41.75	40.95	42.56	4	395.18	389.29	401.15	8
Chinese	40.37	38.79	42.02	5	369.20	357.92	380.85	9
Filipino	39.78	38.35	41.27	6	367.03	356.06	378.30	10
Hispanic ^a	39.19	38.36	40.03	7	409.76	403.46	416.15	7
Black	39.17	38.65	39.69	8	704.18	696.30	712.15	1
Hawaiian	34.68	32.03	37.60	9	463.90	437.03	492.57	5
Alaska Native ^b	28.38	25.17	31.96	10	477.87	430.26	530.45	4
Amer Indian (excl AK) ^c	24.30	22.19	26.62	11	260.08	242.03	279.35	11
All races	45.53	45.35	45.71		595.91	593.94	597.89	
Colon and rectum								
Japanese	9.71	9.03	10.44	1	81.33	76.35	86.62	2
Chinese	7.14	6.49	7.86	2	59.38	54.77	64.36	7
Asian/Pacific Islander	6.88	6.58	7.19	3	60.64	58.35	63.02	6
Alaska Native ^b	6.51	5.08	8.32	4	100.29	78.65	127.28	1
White (total)	6.29	6.22	6.36	5	72.62	71.86	73.39	5
Non-Hispanic white ^a	6.27	6.20	6.35	6	72.97	72.11	73.84	4
Filipino	5.55	5.05	6.10	7	47.57	43.65	51.79	10
Hispanic ^a	4.92	4.67	5.19	8	47.70	45.56	49.94	9
Black	4.51	4.35	4.68	9	76.06	73.42	78.78	3
Hawaiian	4.47	3.66	5.57	10	55.52	46.73	66.17	8
Amer Indian (excl AK) ^c	2.83	2.16	3.70	11	29.31	23.38	36.53	11
All races	6.15	6.09	6.21		72.13	71.43	72.83	
Lung and bronchus								
Hawaiian	9.15	8.01	10.55	1	110.94	98.42	125.28	2
Black	8.83	8.61	9.05	2	139.63	136.25	143.09	1
Non-Hispanic white ^a	8.72	8.63	8.81	3	97.11	96.15	98.08	4
White (total)	8.49	8.41	8.57	4	94.18	93.34	95.03	5
Chinese	8.31	7.64	9.04	5	67.60	62.82	72.74	6
Asian/Pacific Islander	7.30	7.00	7.60	6	64.08	61.77	66.48	7
Filipino	6.98	6.45	7.55	7	63.18	58.69	67.96	8
Alaska Native ^b	6.80	5.40	8.55	8	101.12	80.26	127.14	3
Japanese	6.72	6.16	7.33	9	54.88	50.80	59.27	9
Hispanic ^a	5.81	5.53	6.11	10	53.49	51.18	55.89	10
Amer Indian (excl AK) ^c	3.26	2.60	4.11	11	34.14	27.99	41.54	11
All races	8.43	8.36	8.50		95.38	94.60	96.17	
Prostate								
Non-Hispanic white ^a	17.35	17.21	17.48	1	188.54	187.16	189.92	2
Japanese	16.78	15.79	17.83	2	123.72	117.31	130.45	4
White (total)	16.57	16.46	16.69	3	179.11	177.91	180.31	3
Black	14.79	14.48	15.11	4	239.32	234.52	244.22	1
Hispanic ^a	13.52	13.05	14.01	5	118.87	115.30	122.54	5
Filipino	13.31	12.48	14.20	6	103.62	97.69	109.85	6
Asian/Pacific Islander	12.27	11.83	12.73	7	96.40	93.33	99.55	8
Amer Indian (excl AK) ^c	9.13	7.79	10.69	8	82.51	71.86	94.48	9
Hawaiian	8.96	7.44	10.81	9	96.98	83.60	112.40	7
Chinese	8.74	7.99	9.55	10	65.61	60.64	70.96	10
Alaska Native ^b	4.17	2.99	5.74	11	62.27	44.79	85.44	11
All races	16.22	16.12	16.33		178.39	177.29	179.50	
Urinary bladder								
Non-Hispanic white ^a	3.72	3.66	3.78	1	41.55	40.91	42.21	1
White (total)	3.68	3.63	3.73	2	40.81	40.24	41.39	2
Japanese	2.27	1.93	2.68	3	17.77	15.38	20.48	5
Hispanic ^a	2.24	2.06	2.43	4	20.19	18.75	21.72	3
Chinese	2.16	1.82	2.58	5	17.36	14.93	20.18	6
Asian/Pacific Islander	1.77	1.62	1.94	6	15.06	13.91	16.29	7
Black	1.20	1.12	1.29	7	19.77	18.42	21.22	4
Filipino	1.18	0.98	1.44	8	10.62	8.84	12.70	10
Hawaiian	1.00	0.56	1.83	9	10.91	6.87	17.19	9
Alaska Native ^b	0.90	0.45	1.75	10	14.27	7.09	27.45	8
Amer Indian (excl AK) ^c	0.64	0.37	1.16	11	6.51	3.99	10.39	11
All races	3.30	3.26	3.35		37.25	36.75	37.76	

LL: lower limit of 95% confidence interval; UL: upper limit of 95% confidence interval.

^a Based on data for all Surveillance, Epidemiology, and End Results (SEER) areas except the state of Connecticut, which has a high proportion of patients for which Hispanic ethnicity is unknown.

^b Alaska Native: American Indian/Aleut/Eskimo residents of Alaska at the time of diagnosis.

^c Amer Indian (excl AK): American Indian/Aleut/Eskimo residents of SEER areas, excluding the state of Alaska.

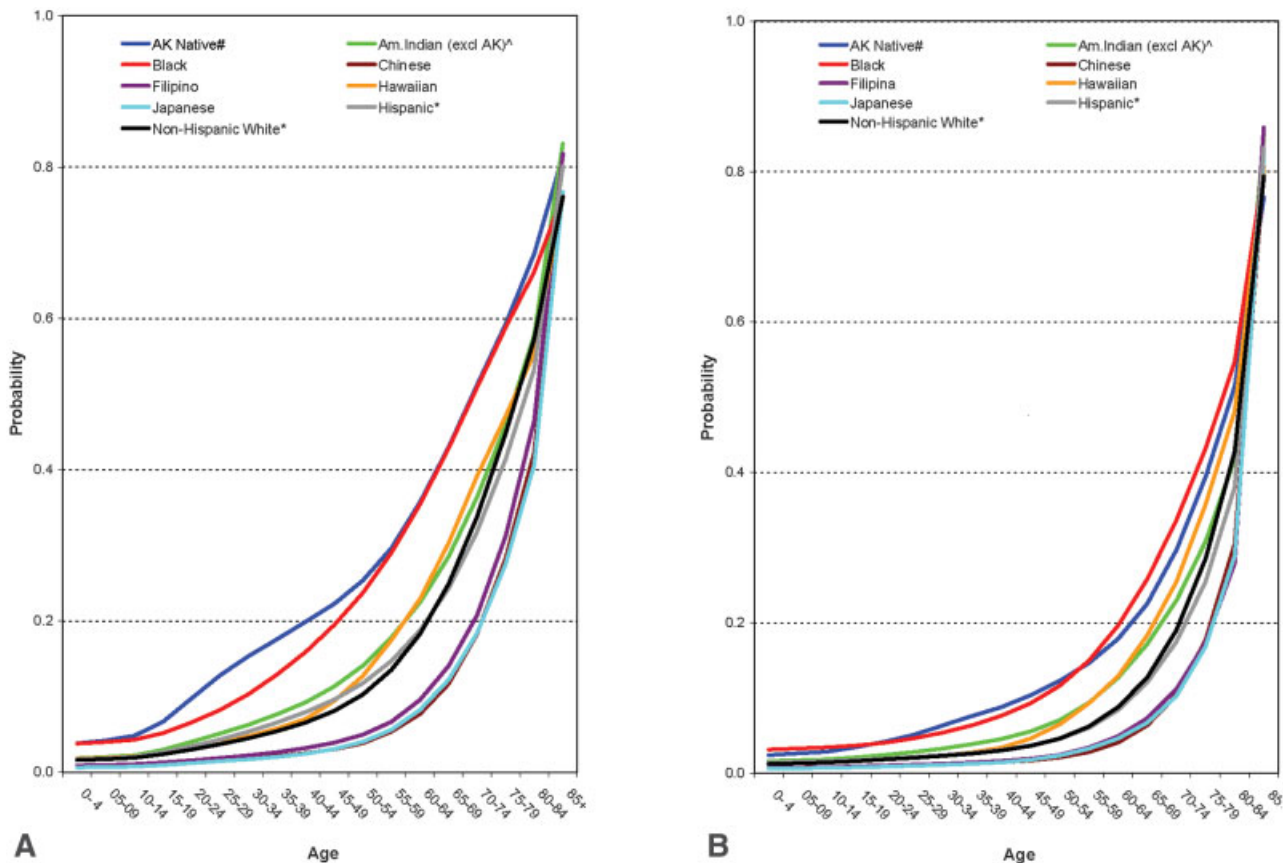


FIGURE 1. (A) The cumulative probability of dying from causes other than cancer by the end of an age interval is illustrated for men in the United States, 1988–1992. (B) The cumulative probability of dying from causes other than cancer by the end of age interval is illustrated for women in the United States, 1988–1992. The asterisk indicates that the results were based on data for 1989–1991 only (CT, Washington DC, LA, ME, MD, MS, NH, NY, ND, OK, VT, and VA were excluded); Alaska Native: American Indian/Aleut/Eskimo residents of Alaska at time of death; Am. Indian (excl AK): American Indian/Aleut/Eskimo residents of the United States, excluding the state of Alaska.

adjusted incidence rate for all cancers combined, yet their lifetime probability of developing cancer is relatively low, ranking eighth among the racial/ethnic groups. A similar pattern is evident in this group for cancers of the colon and rectum, prostate, and urinary bladder. The large difference between the rankings for cancer incidence rate and the lifetime risk may be attributed in part to the high mortality because of other (i.e., noncancer) causes among black men, particularly in younger age groups (Fig. 1A). This results in a smaller proportion of the black male population surviving to older ages, when cancer becomes more prevalent. A similar pattern of high incidence rates relative to low rankings for the risk of developing cancer occurs for black women and for Alaska Native men. These groups also experience high mortality because of other causes, thereby reducing their chances of surviving to older ages and developing cancer (Fig. 1A,B).

Age-Conditional Probability of Developing Cancer

The probabilities of developing selected cancers within the next 10 years are presented for men and women age 60 years in Figure 2. The important thing to note about these probabilities is that they reflect the chances of a cancer diagnosis among the population that has already lived to age 60 years without a cancer diagnosis and, thus, are quite different from the lifetime probabilities of developing cancer. The broad 95% CIs for the risk measures in Alaska Natives and native Hawaiians are because of the smaller population sizes of these racial/ethnic groups. There is less racial/ethnic variation in the age-conditional risks for colorectal cancer than for the other cancers shown. For lung cancer, the racial/ethnic groups appear to separate into two clusters with higher risks among white, black, native Hawaiian, and Alaska Native men and women and with lower risks in the Asian, American Indian, and Hispanic populations. The age-con-

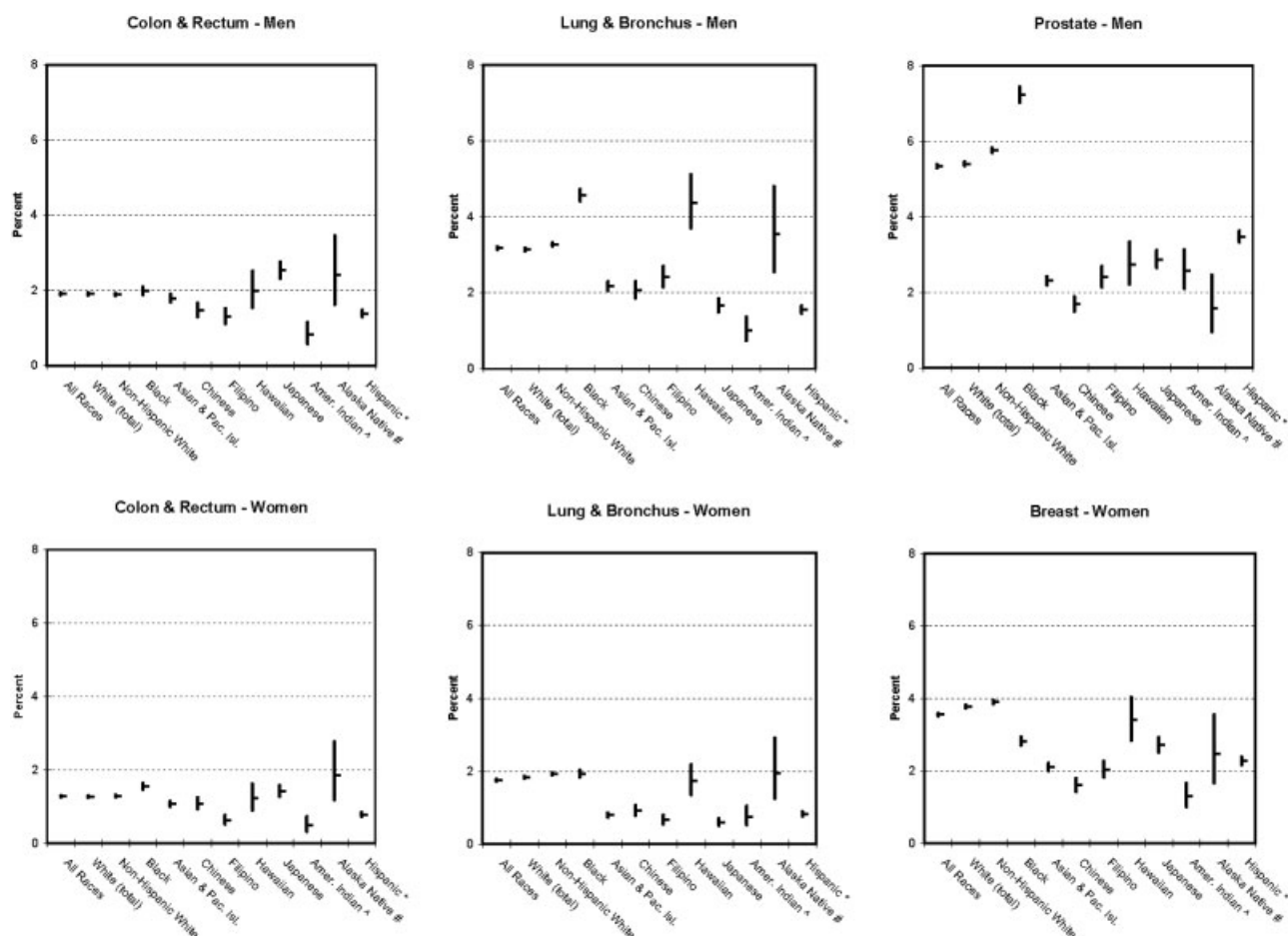


FIGURE 2. The risks (percentages) and 95% confidence intervals of developing invasive cancer between ages 60–70 years are illustrated by racial/ethnic group and gender for the Surveillance, Epidemiology, and End Results (SEER) areas, 1988–1992. The asterisk indicates that data were based on all SEER areas except the state of Connecticut, which has a high proportion of individuals with unknown Hispanic ethnicity. Alaska Native: American Indian/Aleut/Eskimo residents of Alaska at the time of diagnosis; Amer. Indian (excl AK): American Indian/Aleut/Eskimo residents of SEER areas, excluding the state of Alaska.

ditional risks for prostate cancer are highest for black men (7.23%) and non-Hispanic white men (5.76%). Although black men age 60 years clearly have the highest probability of developing prostate cancer within the next 10 years, their lifetime risk of developing this cancer (14.79%) is lower compared with the lifetime risk for non-Hispanic white men (17.35%) and Japanese men (16.78%) (Table 2).

Probability of Dying from Cancer

Tables 3 and 4 shows lifetime probabilities of dying from all cancers combined and for selected cancers, along with age-adjusted mortality rates, by racial/ethnic group among women and men, respectively. Men in the white, native Hawaiian, black, Chinese, and Japanese populations and Alaska Native women all have among the highest lifetime risks of dying from cancer, ranging from approximately 23.3% to 23.9%.

The risk estimates among men in the top six racial/ethnic groups are very nearly the same in magnitude, however, and all of their 95% CIs are overlapping, so the individual rankings for these groups are not very informative. American Indian, Hispanic, and Filipino men and women and Alaska Native men have relatively lower risks of dying from cancer. The risk for all cancers combined is slightly lower for women than for men; however, women have a higher risk of dying from colorectal cancer. The colorectal cancer risks in women exceed those for men in Alaska Native, white, black, and American Indian populations. It also was found that the lifetime risk of dying from cancer of the gallbladder (results not shown) was highest among American Indian women (0.37%; 95% CI, 0.27–0.52%). Black men have the highest lifetime risk of dying from prostate cancer as a result of their exceedingly high mortality rate from this disease. The level of their risk

TABLE 3

Lifetime Risk (Percentage) among Women of Dying from Cancer and Age-Adjusted Mortality Rates by Cancer Type and Racial/Ethnic Group: United States, 1988–1992

Disease type/group	Lifetime risk of dying				Age-adjusted mortality			
	Percent	LL (%)	UL (%)	Rank	Rate	LL	UL	Rank
All invasive cancers								
Alaska Native ^a	23.42	20.36	26.92	1	1	227.81	196.99	262.99
Non-Hispanic White ^b	20.66	20.60	20.71	2	4	174.24	173.74	174.74
White (total)	20.48	20.44	20.52	3	5	172.89	172.56	173.23
Hawaiian	19.68	17.99	21.57	4	3	179.50	165.34	194.77
Chinese	19.45	18.60	20.34	5	7	112.12	108.07	116.29
Black	19.37	19.26	19.47	6	2	204.64	203.49	205.79
Japanese	17.93	17.13	18.76	7	8	110.40	106.00	114.98
Asian/Pacific Islander	17.38	16.94	17.84	8	10	107.24	105.16	109.36
Hispanic ^b	16.84	16.54	17.15	9	6	115.32	113.54	117.12
Amer. Indian (excl AK) ^c	15.39	14.69	16.14	10	9	108.22	104.22	112.35
Filipina	14.16	12.79	15.69	11	11	77.88	73.90	82.12
All races	20.24	20.21	20.28			174.63	174.31	174.95
Colon and rectum								
Alaska Native ^a	4.20	2.69	6.36	1	1	35.95	23.21	53.87
Chinese	2.76	2.40	3.16	2	6	14.27	12.82	15.87
Japanese	2.73	2.39	3.11	3	5	15.76	14.09	17.61
Non-Hispanic White ^b	2.66	2.64	2.68	4	3	20.27	20.11	20.44
White (total)	2.66	2.64	2.67	5	4	20.31	20.20	20.43
Black	2.64	2.60	2.68	6	2	26.18	25.76	26.60
Asian/Pacific Islander	2.21	2.04	2.41	7	8	12.06	11.34	12.82
Hispanic ^b	1.95	1.83	2.07	8	9	12.00	11.42	12.62
Hawaiian	1.90	1.25	2.85	9	7	13.67	9.67	18.98
Amer. Indian (excl AK) ^c	1.77	1.50	2.09	10	10	10.81	9.54	12.22
Filipina	1.56	1.08	2.25	11	11	7.50	6.24	9.06
All races	2.64	2.63	2.66			20.72	20.61	20.83
Lung and bronchus								
Alaska Native ^a	5.47	4.00	7.51	1	1	54.71	40.61	73.30
Hawaiian	4.81	4.01	5.84	2	2	44.97	38.11	52.97
Chinese	4.53	4.10	5.01	3	6	24.82	22.91	26.88
Non-Hispanic White ^b	4.34	4.31	4.36	4	3	38.09	37.86	38.32
White (total)	4.24	4.23	4.26	5	4	37.19	37.03	37.34
Black	3.36	3.32	3.41	6	5	36.26	35.78	36.74
Asian/Pacific Islander	3.15	2.95	3.36	7	8	18.78	17.90	19.70
Amer. Indian (excl AK) ^c	2.48	2.23	2.7	8	7	20.09	18.42	21.89
Japanese	2.36	2.07	2.68	9	9	15.37	13.81	17.11
Filipina	2.31	1.75	3.05	10	11	12.39	10.83	14.23
Hispanic ^b	2.27	2.15	2.39	11	10	14.93	14.29	15.60
All races	4.11	4.09	4.13			36.71	36.57	36.86
Breast								
Non-Hispanic White ^b	3.65	3.62	3.67	1	2	33.10	32.88	33.32
White (total)	3.60	3.58	3.62	2	3	32.71	32.56	32.86
Black	3.31	3.27	3.36	3	1	37.68	37.19	38.18
Hawaiian	2.95	2.30	3.83	4	4	28.41	23.14	34.81
Hispanic ^b	2.39	2.29	2.51	5	5	19.66	18.96	20.39
Chinese	1.91	1.66	2.20	6	11	13.57	12.25	15.01
Japanese	1.81	1.59	2.07	7	7	14.61	13.12	16.27
Asian/Pacific Islander	1.77	1.64	1.90	8	8	14.04	13.37	14.76
Amer. Indian (excl AK) ^c	1.71	1.48	1.97	9	9	13.81	12.46	15.30
Filipina	1.65	1.33	2.14	10	10	13.73	12.33	15.38
Alaska Native ^a	1.50	0.91	2.69	11	6	18.18	11.37	29.46
All races	3.53	3.52	3.55			32.85	32.71	32.99
Corpus and uterus								
Hawaiian	0.94	0.57	1.59	1	1	8.73	5.70	13.06
Black	0.72	0.70	0.74	2	2	7.27	7.05	7.49
Non-Hispanic White ^b	0.50	0.49	0.51	3	3	4.02	3.95	4.10
White (total)	0.50	0.49	0.50	3	3	4.02	3.97	4.07
Hispanic ^b	0.49	0.44	0.55	4	4	3.21	2.92	3.53
Japanese	0.32	0.23	0.46	5	8	2.22	1.65	2.96
Amer. Indian (excl AK) ^c	0.32	0.23	0.46	5	6	2.38	1.83	3.08
Asian/Pacific Islander	0.32	0.27	0.39	5	7	2.29	2.01	2.61
Chinese	0.32	0.24	0.44	5	5	2.47	1.95	3.13
Filipina	0.16	0.11	0.45	6	9	1.37	0.97	2.07
Alaska Native ^a	—	—	—	—	—	—	—	—
All races	0.52	0.51	0.52			4.27	4.23	4.32

LL: lower limit of 95% confidence interval; UL: upper limit of 95% confidence interval.

^a Based on data for 1989–1991 only (CT, Washington DC, LA, ME, MD, MS, NH, NY, ND, OK, VT, and VA excluded).

^b Alaska Native: American Indian/Aleut/Eskimo residents of Alaska at the time of death.

^c Amer Indian (excl AK): American Indian/Aleut/Eskimo residents of the United States, excluding the state of Alaska (no cases).

TABLE 4
Lifetime Risk (Percentage) among Men of Dying from Cancer with Age-Adjusted Mortality Rates by Cancer Type
and Racial/Ethnic Group: United States, 1988–1992

Disease type/group	Lifetime risk of dying				Age-adjusted mortality			
	Percent	LL (%)	UL (%)	Rank	Rate	LL	UL	Rank
All invasive cancers								
Non-Hispanic White ^a	23.94	23.89	24.00	1	3	273.23	272.48	273.98
Hawaiian	23.82	21.94	25.91	2	5	261.69	241.99	283.07
Black	23.80	23.70	23.91	3	1	394.76	392.72	396.82
White (total)	23.61	23.57	23.65	4	4	271.08	270.57	271.59
Chinese	23.37	22.57	24.20	5	7	179.52	173.72	185.51
Japanese	23.29	22.35	24.27	6	8	171.88	165.09	178.92
Asian/Pacific Islander	21.70	21.29	22.11	7	9	170.97	167.96	174.02
Hispanic ^a	19.86	19.55	20.17	8	6	180.38	177.62	183.19
Alaska Native ^b	18.56	16.40	21.00	9	2	280.63	243.45	322.97
Filipino	18.26	17.55	18.99	10	11	137.32	132.15	142.66
Amer. Indian (excl AK) ^c	16.93	16.26	17.63	11	10	163.25	156.99	169.75
All races	23.56	23.52	23.60			279.13	278.64	279.62
Colon and rectum								
Japanese	3.43	3.05	3.85	1	6	25.63	23.11	28.40
Chinese	2.94	2.61	3.31	2	7	21.27	19.21	23.53
Non-Hispanic White ^a	2.64	2.62	2.67	3	3	30.16	29.91	30.42
White (total)	2.63	2.61	2.64	4	4	30.13	29.96	30.31
Asian/Pacific Islander	2.50	2.35	2.67	5	8	19.20	18.18	20.27
Alaska Native ^b	2.36	1.53	3.56	6	1	36.95	23.70	56.10
Hawaiian	2.22	1.70	3.05	7	5	26.44	20.59	34.09
Black	2.14	2.11	2.18	8	2	36.26	35.63	36.91
Filipino	2.07	1.82	2.36	9	10	15.24	13.55	17.12
Hispanic ^a	2.01	1.91	2.13	10	9	18.12	17.24	19.03
Amer. Indian (excl AK) ^c	1.48	1.27	1.73	11	11	14.10	12.28	16.16
All races	2.57	2.55	2.58			30.41	30.24	30.57
Lung and bronchus								
Hawaiian	8.52	7.37	9.93	1	2	93.89	82.44	107.06
Non-Hispanic White ^a	7.98	7.95	8.02	2	3	89.58	89.16	89.99
White (total)	7.78	7.75	7.80	3	4	87.75	87.46	88.03
Black	7.75	7.68	7.81	4	1	123.91	122.81	125.02
Chinese	6.81	6.35	7.30	5	6	51.84	48.75	55.13
Asian/Pacific Islander	5.59	5.38	5.81	6	9	44.80	43.30	46.36
Alaska Native ^b	5.57	4.35	7.11	7	5	81.55	62.67	105.70
Japanese	5.38	4.92	5.89	8	10	40.67	37.48	44.11
Hispanic ^a	4.93	4.77	5.10	9	8	45.39	44.03	46.80
Amer. Indian (excl AK) ^c	4.84	4.50	5.22	10	7	48.99	45.76	52.43
Filipino	4.46	4.11	4.83	11	11	36.12	33.55	38.87
All races	7.72	7.70	7.74			89.89	89.61	90.16
Prostate								
Black	4.24	4.19	4.29	1	1	75.96	74.99	76.95
Filipino	3.58	3.20	4.00	2	7	21.86	19.67	24.23
Hawaiian	3.29	2.34	4.59	3	4	29.48	22.04	39.02
Non-Hispanic White ^a	3.14	3.12	3.17	4	2	36.00	35.71	36.29
White (total)	3.07	3.05	3.09	5	3	35.44	35.25	35.64
Hispanic ^a	3.02	2.87	3.18	6	5	24.48	23.36	25.63
Japanese	2.77	2.37	3.21	7	8	17.76	15.41	20.40
Amer. Indian (excl AK) ^c	2.75	2.42	3.11	8	6	23.73	21.10	26.62
Asian/Pacific Islander	2.59	2.41	2.79	9	9	16.84	15.76	17.98
Chinese	1.67	1.39	2.00	10	11	10.48	8.89	12.30
Alaska Native ^b	0.93	0.41	1.87	11	10	14.19	6.08	29.03
All races	3.20	3.19	3.22			38.21	38.02	38.41
Urinary bladder								
Non-Hispanic White ^a	0.72	0.71	0.73	1	1	8.21	8.07	8.34
White (total)	0.71	0.70	0.72	2	2	8.16	8.07	8.26
Hispanic ^a	0.53	0.47	0.59	3	5	4.41	3.95	4.90
Hawaiian	0.46	0.18	1.15	4	4	4.36	2.05	8.92
Japanese	0.45	0.30	0.66	5	7	2.98	2.09	4.16
Chinese	0.43	0.30	0.60	6	8	2.88	2.13	3.86
Asian/Pacific Islander	0.40	0.33	0.48	7	9	2.75	2.34	3.21
Black	0.37	0.36	0.39	8	3	6.56	6.28	6.85
Amer. Indian (excl AK) ^c	0.26	0.17	0.39	9	10	2.31	1.56	3.35
Filipino	0.23	0.15	0.34	10	11	1.60	1.08	2.31
Alaska Native ^b	0.21	0.04	0.83	11	6	3.25	0.62	12.80
All races	0.67	0.66	0.67			7.93	7.84	8.02

LL: lower limit of 95% confidence interval; UL: upper limit of 95% confidence interval.

^a Based on data for 1989–1991 only (CT, Washington DC, LA, ME, MD, MS, NH, NY, ND, OK, VT, and VA excluded).

^b Alaska Native: American Indian/Aleut/Eskimo residents of Alaska at the time of death.

^c Amer Indian (excl AK): American Indian/Aleut/Eskimo residents of the United States, excluding the state of Alaska (no cases).

is attenuated relative to the remaining racial/ethnic groups, however, because of high mortality from other causes among black men. The low rankings for the risk of dying from cancer relative to the (high) rankings for the age-adjusted mortality rates in Alaska Native men and black women is similar to the pattern seen for the lifetime risk of developing cancer and the incidence rates in these groups.

DISCUSSION

Racial/ethnic patterns are described for the lifetime and age-conditional probabilities of developing cancer and for the probabilities of dying from cancer. Ninety-five percent CI calculations, which are included now with the DevCan program, aid in interpreting these cancer risk estimates and indicate the increased statistical variability associated with point estimates for smaller racial/ethnic populations. The range in the cancer risk estimates across specific Asian populations indicates that important differences can be obscured when analyses are based on a broad racial/ethnic category, such as Asian/Pacific Islander. Furthermore, Alaska Native cancer risks differed markedly from those for American Indians in the remaining geographic areas for cancers of the colon and rectum, lung and bronchus, prostate, and female breast. It should be noted that approximately 46% of the American Indians who were diagnosed with cancer during the period of the current study derived from the New Mexico Tumor Registry, and 29% derived from the Seattle/Puget Sound Tumor Registry. Therefore, our cancer risk estimates for American Indians are influenced heavily by southwest and Pacific northwest tribes and should not be considered representative of all American Indian tribes in the continental United States. Cancer rates among American Indians in the southwest, in particular, reportedly are lower than the rates among tribes in other regions.^{23,24}

The risk estimates for developing cancer (Tables 1, 2) are based on data from the SEER regions, which covered approximately 14% of the United States population during the period of the current study. The risk estimates for dying from cancer (Tables 3, 4), conversely, are based on data for the entire United States population. This difference in geographic coverage may account for some of the differences in the racial/ethnic risk rankings between Tables 1 and 2 and Tables 3 and 4. Our rationale for choosing these population bases was to include the most complete racial/ethnic cancer incidence and mortality data available, although the geographic coverage differs between the two.

A few patterns of lifetime risk seem counterintuitive. For example, although black men have high over-

all cancer incidence rates, they experience a relatively low lifetime probability of actually developing cancer. This type of discrepancy can result when mortality from causes other than cancer is particularly high, which leads to a smaller proportion of the cohort reaching older ages, when cancers predominantly occur. In this example, the cancer incidence rates present an incomplete picture of the overall cancer burden experienced by black men. Conversely, Japanese men and women experience relatively low non-cancer mortality; thus, a greater proportion of this cohort survives to older ages, when cancer rates are higher. Thus, the entire spectrum of diseases and their interactions must be considered when evaluating racial/ethnic health disparities.

It also is important to recognize factors that may have an impact on the cross-sectional incidence rates used to calculate DevCan lifetime risk estimates. For example, the introduction and rapid diffusion of testing for prostate-specific antigen (PSA) in the late 1980s and early 1990s paralleled dramatic increases in prostate cancer incidence rates.²⁵⁻³⁰ An examination of the DevCan lifetime risk estimates for prostate cancer, calculated from cross-sectional incidence rates during that period, shows risk estimates for white men that initially were lower than those for black men (Fig. 3).³¹ This is followed by a period during the late 1980s and early 1990s (as PSA testing and incidence rates began to rise), when the DevCan lifetime risk estimates increase dramatically, with the risk for white men temporarily exceeding that for black men. This result is spurious, at least in part, because DevCan risk estimates are reflecting incidence rates that initially were rising more quickly among white men than among black men.^{26,27,29} Conversely, if we were able to follow an actual cohort of individuals, many of the men who are diagnosed with prostate cancer after a positive PSA test otherwise may not have been diagnosed until some later point in their lives, after they developed clinical symptoms. Thus, the lifetime risk for this cohort would shift toward the younger ages, but the overall lifetime risk may be expected to show little or no change. (Some portion of screen-detected, slow-growing tumors may never have been detected clinically before the patient died from other causes of death.) Because the DevCan lifetime risk estimates are derived by using cross-sectional incidence and mortality rates, they must be interpreted with caution when events, such as the widespread and rapid implementation of a new screening modality, have an impact on disease rates.

Age-conditional risk estimates may be more meaningful to an individual who wants to know how published cancer risk statistics apply to them⁵; and,

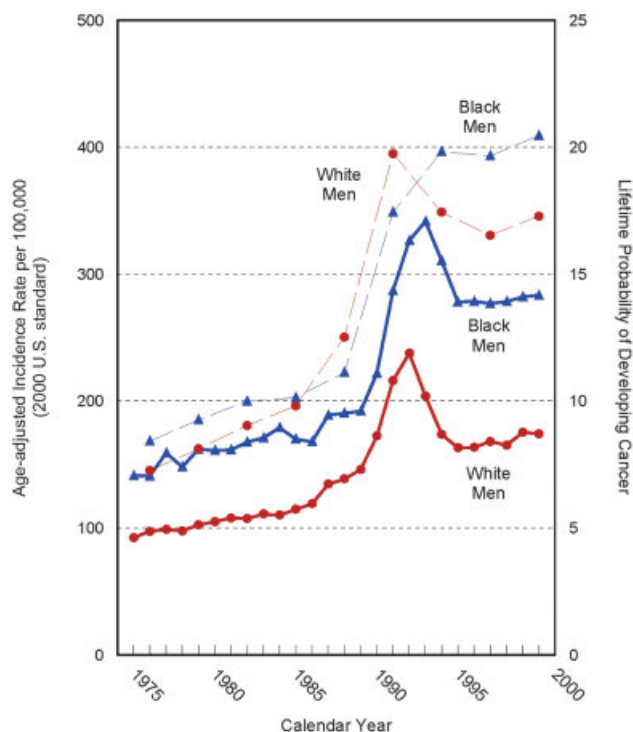


FIGURE 3. Age-adjusted incidence rates for invasive cancer of the prostate (solid lines) and the lifetime probability of developing prostate cancer (dashed lines) are illustrated for black men and white men. The 1975–2000 time trend analyses in this figure were based on data from the following Surveillance, Epidemiology, and End Results regions: metropolitan Atlanta, Detroit, San Francisco/Oakland, Seattle/Puget Sound, and the states of Connecticut, Hawaii, Iowa, New Mexico, and Utah [(see National Cancer Institute, Surveillance, Epidemiology, and End Results (SEER) Program. DevCan database: “SEER 9 incidence and mortality, 1975–2000, follow-back year_1973” (released April 2003 based on the November 2002 submission). Bethesda: National Cancer Institute, DCCPS, Surveillance Research Program, Cancer Statistics Branch, 2003. Underlying mortality data provided by NCHS. Available from URL: <http://www.cdc.gov/nchs> [accessed October 7, 2004]³¹].

when more is known about individual risk factors, more sophisticated models can provide even better estimates of cancer risk.³² Age-conditional risks also may reflect the immediate future of an individual better; because, unlike the lifetime risk estimates, age-conditional risks do not require an assumption that the underlying cross-sectional incidence and mortality rates remain constant over an entire lifetime.

The impact of racial/ethnic misclassification must be considered when interpreting these results. Information on race/ethnicity is obtained routinely from medical records, death certificates, or surname lists, and errors in classification as well as noncomparability with self-reported data from the decennial Census are well documented.^{33–39} Racial/ethnic misclassification on death certificates reportedly reduces the nu-

merators (deaths) for specific racial/ethnic groups.³⁷ However, when combined with the under coverage of specific racial/ethnic groups in the United States Census (population denominators), and when it was assumed that misclassification is uniform across age groups, Rosenberg et al. reported that the expected net impact on age-adjusted United States mortality rates was a slight overestimate (1%) of the white mortality rate and a larger overestimate (5%) of mortality in the black population; whereas mortality rates for American Indians (including Alaska Natives) and Asians/Pacific Islanders were underestimated substantially (nearly 21% and 11%, respectively), and those for Hispanics were underestimated slightly (about 2%). To investigate the potential effect of racial/ethnic misclassification in our study, we assumed that similar patterns of misclassification held for the cancer mortality and incidence rates that were used to generate the DevCan lifetime risk estimates. We then produced a set of “adjusted” lifetime risk estimates that took into account racial/ethnic misclassification (not shown). The revised lifetime risks for dying from cancer among white, black, and Hispanic men showed no meaningful changes (from 23.61% [Table 4] to 23.57% [revised]; from 23.80% to 23.83%; and from 19.86% to 19.89%, respectively), whereas the risk estimates for American Indians and Asians/Pacific Islanders increased slightly (from 16.93% to 17.08% and from 21.70% to 22.20%, respectively). These findings suggest that racial/ethnic misclassification of the magnitude found by Rosenberg et al. does impact the DevCan lifetime risk estimates, but the differences are small when it is assumed that misclassifications affect both mortality and incidence rates. Ongoing efforts to cross-reference cancer registry files systematically with administrative records from the Indian Health Service will help to reduce the misclassification of American Indians.³⁵

In summary, the lifetime and age-conditional probabilities of developing cancer, the probabilities of dying from cancer, and the associated confidence intervals now may be calculated for several more racial/ethnic groups by using newly available data bases with current versions of the DevCan software.⁹ Data bases centered on the 1990 Census produce these statistics for Alaska Natives (literally, for American Indian/Aleut/Eskimo populations in Alaska), American Indian/Aleut/Eskimo (excluding Alaskans), black, Chinese, Filipino, native Hawaiian, Japanese, white (total, non-Hispanic), and Hispanic populations. Alternative data bases with more recent incidence and mortality data are available for 1992 and forward, but those are limited to fewer racial/ethnic groups (American Indian/Alaska Native, Asian/Pacific Islander, black, white, and

Hispanic). This is because detailed population estimates, which are needed for rate calculations, are limited to those broader groups in non-Census years. The data bases with broader racial/ethnic groups, although they were not used in the current analysis, can be accessed online.⁹ Another option is to develop a data base centered on the 2000 decennial Census in a manner similar to our analysis, which was centered on the 1990 Census. However, problems concerning the noncomparability of race information from different sources now are compounded, because the 2000 Census tabulated multiple-race responses from individuals of mixed heritage, and most hospital records and death certificates contain only single-race information. Methods for “bridging” the multiple-race population denominator data to single-race categories, to make them more comparable to the numerator data, are based on statistical models that have been developed only for major race groups (white, black, Asian/Pacific Islander, American Indian/Aleut/Eskimo).⁴⁰ Even if criteria were developed for reassigning multiple-race populations to single-race groups, this would not be possible for all geographic areas because of the Census Bureau’s disclosure policy, which prevents the release of age-specific, race-specific, gender-specific population counts < 100.⁴¹

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